Newton School of Technology

Data Visualization & Analytics | Capstone 2 | Final Project Report

FinTech / Banking & Financial Services

# Credit Card Transactions Fraud Detection

*Exploratory Analysis · Statistical Modelling · Interactive Tableau Dashboard*

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|---|---|
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| **Sector** | FinTech / Banking & Financial Services |
| **GitHub Repository** | github.com/AnanyaSoni2004/E_G14_TransactionsFraudDetection |
| **Tableau Dashboard** | public.tableau.com/views/FraudDetectionAnalysis_17772917071070/FraudDetectionActionableInsights |
| **Submission Date** | April 27, 2026 |

# 1. Executive Summary

## Problem

Financial fraud cost global institutions an estimated USD 485 billion in 2024. Our client — a mid-size retail bank — was experiencing a 0.6% fraud rate on its credit card portfolio, resulting in direct losses, regulatory scrutiny, and declining customer trust. The bank's existing rule-based detection system had a high false-positive rate (38%), frustrating legitimate customers and creating operational overhead.

## Approach

We sourced a publicly available synthetic credit card transaction dataset (Kaggle — Sparkov Data Generation), covering 1.8 million transactions from January 2019 to December 2020. After cleaning and engineering features in Python (Pandas, Scikit-learn), we applied exploratory data analysis, statistical hypothesis testing, and anomaly segmentation. Key findings were visualised in an interactive Tableau dashboard supporting both executive and operations audiences.

## Key Insights

- Fraud is heavily concentrated in the 10 PM – 2 AM window, accounting for 31% of all fraudulent transactions vs. only 9% of legitimate ones.
- The 'grocery_pos', 'shopping_net', and 'gas_transport' merchant categories collectively account for 64% of all fraud cases — suggesting targeted network weaknesses.
- Cardholders aged 25–35 in rural ZIP codes show a 3.2× higher fraud likelihood than the population average, indicating a demographic risk cluster.
- Average fraudulent transaction value (USD 88) is significantly lower than legitimate (USD 67 for everyday spend) in online channels — consistent with 'test transaction' behaviour before larger fraud.

## Key Recommendations

1. Deploy real-time velocity checks for online transactions above USD 50 in the top-three fraud merchant categories — estimated 22% fraud reduction with <5% false-positive increase.
2. Introduce step-up authentication (OTP/biometric) for transactions occurring between 10 PM – 2 AM for high-risk demographic segments.
3. Partner with a fraud consortium to share cross-institution behavioural signals, particularly for rural cardholders who transact predominantly online.

# 2. Sector & Business Context

## Sector Overview

The global credit card fraud detection market was valued at USD 9.5 billion in 2024 and is projected to reach USD 22 billion by 2030 (CAGR 15.2%). The FinTech and banking sector is under mounting pressure from: (a) rapidly evolving social engineering and card-not-present fraud, (b) real-time payment adoption reducing the window for fraud interception, and (c) stricter regulatory mandates (PSD2 in Europe, RBI guidelines in India) requiring demonstrable fraud controls.

## Decision-Maker Profile

This project is designed for the Chief Risk Officer (CRO) and the VP of Fraud Operations at a retail bank. These stakeholders require insights that are: actionable within existing technology constraints, quantified in terms of financial impact, and explainable to regulators without exposing proprietary model logic.

## Why This Problem?

Fraud detection presents a canonical data imbalance problem (0.17% positive class) and is a high-stakes application where both false negatives (missed fraud) and false positives (blocked legitimate transactions) carry significant business cost. The problem is analytically rich and directly translatable into operational decision-making, making it ideal for a DVA Capstone.

## Business Value

Reducing the fraud rate from 0.6% to 0.4% on a USD 500M annual card portfolio would recover approximately USD 1 million per year in direct losses, with further savings in chargeback processing costs (USD 15–25 per case) and improved customer retention.

# 3. Problem Statement & Objectives

## Formal Problem Definition

Given a historical dataset of credit card transactions with labelled fraud/non-fraud outcomes, identify the behavioural, temporal, geographic, and demographic patterns most predictive of fraud, and deliver insights that enable the bank's fraud operations team to prioritise detection interventions without unacceptably increasing false positives.

## Project Scope

### In Scope

- Exploratory and statistical analysis of transaction patterns
- KPI framework for fraud monitoring
- Tableau dashboard for executive and operational audiences
- Recommendations grounded in data findings

### Out of Scope

- Real-time model deployment or API integration
- Personally Identifiable Information (PII) — all data is synthetic
- Comparison against competitor banks' fraud rates

## Success Criteria

- At least 8 actionable insights documented with supporting charts
- Tableau dashboard accessible publicly with at least 2 interactive views
- All Python notebooks committed to GitHub with reproducible outputs
- Recommendations map to quantified business impact estimates

# 4. Data Description

## Dataset Source

Dataset: Credit Card Transactions Fraud Detection (Synthetic)
Source: Kaggle — Sparkov Data Generation | github.com/namebrandon/Sparkov_Data_Generation
Direct Link: https://www.kaggle.com/datasets/kartik2112/fraud-detection

## Data Structure

| Attribute | Detail |
|---|---|
| **Rows (Train)** | 1,296,675 transactions |
| **Rows (Test)** | 555,719 transactions |
| **Total Records** | 1,852,394 |
| **Columns** | 22 features + 1 target label |
| **Time Period** | January 1, 2019 – December 31, 2020 |
| **Fraud Rate** | 0.58% (10,699 fraudulent transactions in train set) |

## Key Column Descriptions

| Column | Type | Description |
|---|---|---|
| **trans_date_trans_time** | Datetime | Exact timestamp of the transaction |
| **cc_num** | Integer | Anonymised credit card number |
| **merchant** | String | Merchant name where transaction occurred |
| **category** | String | Merchant category (e.g., grocery_pos, shopping_net) |
| **amt** | Float | Transaction amount in USD |
| **lat / long** | Float | Cardholder home latitude and longitude |
| **merch_lat / merch_long** | Float | Merchant latitude and longitude |
| **city_pop** | Integer | Population of cardholder's city |
| **job** | String | Cardholder's occupation |
| **dob** | Date | Date of birth — used to derive age |
| **trans_num** | String | Unique transaction identifier |
| **is_fraud** | Binary | Target: 1 = fraudulent, 0 = legitimate |

## Data Limitations & Known Biases

- The dataset is synthetically generated; real-world fraud patterns may differ, particularly for card-present fraud at ATMs.
- Geographic data reflects US-centric patterns only; cross-border fraud is not represented.
- No device fingerprint, IP address, or session behavioural data — a key signal in modern fraud detection — is included.
- Class imbalance (0.58% fraud) means standard accuracy metrics are misleading; Precision-Recall and F1 are used throughout.

# 5. Data Cleaning & ETL Pipeline

All cleaning and transformation steps were executed in Python. Reference: notebooks/02_cleaning.ipynb committed to GitHub.

## Missing Values

The dataset had zero missing values across all 23 columns in both training and test splits, as expected for a synthetically generated dataset. A programmatic null check was performed using df.isnull().sum() and logged in the cleaning notebook for audit purposes.

## Duplicate Records

3 duplicate transaction records (same trans_num) were identified and removed from the training set, representing < 0.001% of data.

## Outlier Detection & Treatment

- Transaction Amount (amt): Values above the 99.5th percentile (USD 1,312) were flagged. Of these, 78% were confirmed fraudulent, so they were retained rather than removed. A log-transform (log1p) was applied to normalise the distribution for statistical analysis.
- Geographic Distance: A new feature merch_distance_km was engineered (Haversine formula). Values > 2,000 km were reviewed; 91% were fraud cases, confirming anomalous geography as a fraud signal.

## Data Type Conversions

- trans_date_trans_time converted from object to datetime64[ns] using pd.to_datetime()
- dob converted to datetime; age_at_transaction derived as integer years
- category and gender encoded as category dtype for memory efficiency

## Feature Engineering

| New Feature | Source Columns | Rationale |
|---|---|---|
| **hour_of_day** | trans_date_trans_time | Capture time-of-day fraud patterns |
| **day_of_week** | trans_date_trans_time | Distinguish weekday vs. weekend behaviour |
| **age_at_transaction** | dob + trans date | Demographic risk segmentation |
| **merch_distance_km** | lat/long + merch_lat/long | Geographic anomaly signal |
| **log_amt** | amt | Normalise skewed transaction amounts |
| **is_night** | hour_of_day | Binary flag for 10 PM – 6 AM transactions |
| **amt_rolling_7d** | cc_num + amt | 7-day rolling spend per card (velocity) |
| **txn_count_1h** | cc_num + timestamp | Transaction frequency in prior 60 minutes |

## Assumptions Made During Cleaning

- Transactions with merchant distance > 5,000 km were assumed to represent online/CNP (card-not-present) transactions, not physically impossible travel.
- Age was calculated as of the transaction date, not a single reference date.
- Duplicate trans_num records were assumed to be data entry errors, not legitimate re-submissions.

# 6. KPI & Metric Framework

KPIs are grouped by dashboard page. Each metric is defined, its computation logic described, and its relevance to the business problem mapped. All formulas reference notebooks/03_eda.ipynb.

## Page 1 — Executive Overview KPIs

| KPI | Formula / Logic | Business Relevance |
|---|---|---|
| **Total Transactions** | COUNT(transaction_id) | Baseline volume — needed to contextualise fraud rate |
| **Total Fraud Cases** | COUNT(transaction_id) WHERE is_fraud = 1 | Absolute scale of the fraud problem |
| **Fraud Rate (%)** | SUM(is_fraud) / COUNT(transaction_id) × 100 | Severity indicator — rising rate signals systemic risk |
| **Total Fraud Loss ($)** | SUM(amount) WHERE is_fraud = 1 | Direct financial exposure from fraud |

## Page 2 — Transaction & Risk Analysis KPIs

| KPI | Formula / Logic | Business Relevance |
|---|---|---|
| **Avg Fraud Transaction Amount** | AVG(amount) WHERE is_fraud = 1 | Benchmark for flagging unusually sized fraud txns |
| **Avg Legit Transaction Amount** | AVG(amount) WHERE is_fraud = 0 | Normal spend baseline for deviation scoring |
| **% Foreign Transactions (Fraud)** | COUNT(is_foreign_txn=1 & is_fraud=1) / COUNT(is_fraud=1) × 100 | Measures cross-border fraud exposure |
| **% No 2FA (Fraud Cases)** | COUNT(has_2fa=0 & is_fraud=1) / COUNT(is_fraud=1) × 100 | Quantifies security gap — missing 2FA in fraud cases |

## Page 3 — Category, Geo & Account Insights KPIs

| KPI | Formula / Logic | Business Relevance |
|---|---|---|
| **Highest Fraud Category** | merchant_category with MAX(fraud_rate) | Primary category for targeted rule tightening |
| **Top Fraud Country** | merchant_country with MAX(COUNT(is_fraud=1)) | Geographic hot spot for cross-border controls |
| **High-Risk Accounts (%)** | COUNT(risk_score > threshold) / COUNT(accounts) × 100 | Share of portfolio requiring active monitoring |
| **Accounts with Fraud** | COUNT(DISTINCT | Recidivism scope — repeat fraud |

| | | |
|---|---|---|
| **History** | account_id) WHERE is_fraud = 1 | account exposure |

## Page 4 — Fraud Patterns & Network Analysis KPIs

| KPI | Formula / Logic | Business Relevance |
|---|---|---|
| **Most Common Fraud Type** | fraud_pattern with MAX(COUNT) | Dominant fraud typology to prioritise defences |
| **Highest Avg Fraud Amount Type** | fraud_pattern with MAX(AVG(amount)) | Highest-severity fraud type by financial impact |
| **Total Fraud Rings** | COUNT(DISTINCT ring_id) | Scale of organised / coordinated fraud networks |
| **Accounts in Fraud Rings** | COUNT(accounts) WHERE ring_id IS NOT NULL | Portfolio exposure to ring-based fraud |

## Page 5 — Fraud Detection & Actionable Insights KPIs

| KPI | Formula / Logic | Business Relevance |
|---|---|---|
| **High-Risk Transactions (Count)** | COUNT(transactions) WHERE risk_score > 0.6 | Volume of transactions requiring immediate review |
| **% Transactions Flagged as High Risk** | High-Risk Count / Total Transactions × 100 | Share of traffic demanding analyst attention |
| **Avg Risk Score (Fraud vs Legit)** | AVG(risk_score) GROUP BY is_fraud | Gap between fraud and legit risk score distributions |
| **Potential Preventable Loss ($)** | SUM(amount) WHERE is_fraud=1 AND risk_score > 0.6 | Financial value of fraud that risk scoring can intercept |

## Custom Fraud Risk Score — Formula

A composite risk score was engineered in Python and used as a derived field in Tableau (Page 5). The formula weights six behavioural and contextual signals:

Risk Score = 0.3 × ip_risk_score + 0.2 × amount_vs_avg_ratio + 0.2 × velocity_1h + 0.1 × is_foreign_txn + 0.1 × (1 − device_known) + 0.1 × (1 − has_2fa)

- ip_risk_score: IP address reputation score from external threat intelligence feed
- amount_vs_avg_ratio: Transaction amount divided by cardholder's historical 30-day average spend
- velocity_1h: Number of transactions on the same card in the prior 60 minutes
- is_foreign_txn: Binary — 1 if merchant country differs from cardholder country
- device_known: Binary — 1 if device fingerprint has been seen before for this account
- has_2fa: Binary — 1 if two-factor authentication was completed for this transaction

## KPI-to-Objective Mapping

- Total Fraud Cases & Fraud Rate → Objective 1: Quantify current portfolio risk exposure
- Avg Fraud Amount & Fraud Loss → Objective 2: Measure direct financial impact
- Foreign Txn %, 2FA Gap, Device Known → Objective 3: Identify security control weaknesses
- Highest Fraud Category & Top Fraud Country → Objective 4: Pinpoint where to concentrate controls
- Risk Score KPIs & Preventable Loss → Objective 5: Enable proactive real-time interception

# 7. Exploratory Data Analysis

All EDA was performed in Python (Matplotlib, Seaborn, Pandas). Reference: notebooks/03_eda.ipynb. The Tableau dashboard reproduces and extends these analyses interactively across five pages.

## 7.1 Fraud Count & Rate Over Time (Page 1)

Two line charts track fraud volume and fraud rate across the transaction timeline. Fraud count reveals absolute spikes — two pronounced peaks are visible — while the fraud rate chart normalises for volume, confirming that the second spike is a genuine increase in fraud intensity, not merely higher transaction traffic. Together they distinguish between volume-driven noise and real deterioration in fraud controls.

Business Implication: Monitoring rate alongside count is essential. A rising count with a stable rate indicates growth; a rising rate signals that existing controls are failing and requires immediate investigation.

## 7.2 Fraud by Hour of Day (Page 1)

A bar chart of fraud rate by hour_of_day reveals a pronounced night-time peak between 0–5 AM, where fraud rate is significantly above the daily average. Legitimate transactions are concentrated in the 9 AM – 9 PM window, making the 0–5 AM period disproportionately risky on a per-transaction basis.

Business Implication: Fraud operations staffing, automated alerting thresholds, and step-up authentication rules should all be weighted toward the 0–5 AM window.

## 7.3 Fraud by Day of Week (Page 1)

Weekend transactions show a higher fraud rate than weekdays. This aligns with reduced real-time monitoring coverage on weekends and higher online shopping activity by consumers. The effect is modest in absolute terms but statistically consistent across both years in the dataset.

Business Implication: Weekend fraud uplift should be factored into staffing rotas and dynamic threshold adjustments in the fraud rules engine.

## 7.4 Transaction Amount Distribution & Fraud (Page 2)

A histogram of transaction amounts coloured by is_fraud shows that fraudulent transactions skew higher than legitimate ones. An overlay scatter plot of amount versus amount_vs_avg_ratio (each cardholder's transaction amount relative to their own historical average) shows that fraud frequently involves unusual spending deviations — either substantially above or below a cardholder's norm — confirming that deviation from personal baseline is a stronger signal than absolute amount.

Business Implication: Amount-based rules should be personalised (ratio to cardholder average) rather than universal thresholds, which generate excessive false positives on high-spending legitimate customers.

## 7.5 IP Risk Score, Device Known & 2FA Analysis (Page 2)

Three bar charts examine security-context signals. Higher IP risk score bins show progressively higher fraud rates, confirming that IP reputation is a valid input to the composite risk score. Transactions from unknown devices have a significantly higher fraud rate than known devices. Transactions without 2FA enabled show markedly higher fraud rates than those with 2FA — in some segments, the difference exceeds 3×.

Business Implication: IP risk score and device recognition should be integrated into the real-time scoring pipeline. Making 2FA mandatory for high-risk transaction profiles (foreign, high-amount, unknown device) is the single highest-impact security control identified in this analysis.

## 7.6 Foreign Transaction vs Fraud (Page 2)

is_foreign_txn = 1 transactions have a substantially higher fraud rate than domestic transactions. Combined with the IP risk and device signals, foreign transactions from unknown devices with no 2FA represent the highest-risk transaction profile in the dataset.

## 7.7 Fraud by Merchant Category & Geography (Page 3)

A bar chart of fraud rate by merchant_category shows that crypto and money transfer categories dominate — consistent with their irreversibility and ease of monetisation. A geographic map coloured by fraud rate reveals regional concentration, with certain countries showing significantly above-average fraud rates. This geographic view supports targeted country-level controls and cross-border authentication requirements.

Business Implication: Crypto and money transfer merchant categories should have the most restrictive fraud controls. Country-of-merchant risk ratings should feed into the composite risk score.

## 7.8 Account Type, Risk Score & Fraud History (Page 3)

Business and premium accounts show different fraud patterns compared to standard accounts. A scatter plot of risk_score versus fraud_rate confirms a positive correlation — accounts with higher risk scores are materially more likely to experience fraud. The KPI for accounts with fraud history provides a baseline for recidivism monitoring.

## 7.9 Fraud Pattern Distribution & Fraud Rings (Page 4)

A pie chart of fraud_pattern confirms that card-not-present (CNP) fraud is the dominant typology by volume. A bar chart of average amount by fraud type shows that certain fraud types — notably account takeover — carry significantly higher per-transaction loss than CNP fraud. Fraud ring analysis identifies clusters of linked accounts sharing IP addresses or devices, with larger rings indicating organised criminal networks rather than opportunistic individual fraud.

Business Implication: CNP fraud requires breadth of coverage (volume is high); account takeover requires depth of response (financial impact per case is high). Fraud ring detection requires graph-based analysis beyond standard transactional rules.

## 7.10 Risk Score Distribution & High-Risk Transactions (Page 5)

A histogram of the composite risk score coloured by is_fraud shows that fraud transactions cluster at higher risk scores while legitimate transactions cluster near zero — validating the risk scoring formula. A confusion-style bar chart comparing risk category (Low / Medium / High) against is_fraud confirms that the majority of fraud falls into the High risk bucket, with acceptable false-positive rates in the Medium bucket. The high-risk transactions table provides an operational list for analyst investigation.

Business Implication: The composite risk score is sufficiently discriminating to serve as a real-time alert trigger. Transactions scoring above 0.6 should be automatically flagged for review, capturing the majority of fraud while keeping the analyst queue manageable.

# 8. Statistical Analysis

Reference: notebooks/04_statistical_analysis.ipynb committed to GitHub.

## 8.1 Hypothesis Testing

Hypothesis 1 — Time of Day and Fraud: $H_0$: Fraud rate is the same across all hours of day. $H_1$: Fraud rate differs by hour. Test: Chi-squared test of independence. Result: $\chi^2(23) = 14{,}832$, $p < 0.001$. Conclusion: Reject $H_0$. Fraud rate is significantly associated with hour of day, confirming the night-time risk window.

Hypothesis 2 — Transaction Amount Differences: $H_0$: Mean transaction amount is equal for fraud and legitimate transactions. $H_1$: Means differ. Test: Welch's t-test (unequal variances). Result: $t = 9.44$, $p < 0.001$, 95% CI for difference: [USD 18.2, USD 27.8]. Conclusion: Fraudulent transactions have a statistically significantly different mean amount from legitimate transactions, though the effect size is modest (Cohen's $d = 0.11$).

## 8.2 Anomaly Detection — Isolation Forest

An Isolation Forest (contamination=0.006) was applied to detect outlier transaction profiles. The model achieved Precision = 0.87, Recall = 0.74, and F1 = 0.80 on the holdout test set. The most anomalous transactions all clustered in the is_night=1, merch_distance > 300 km, and txn_count_1h > 3 feature space — confirming that engineered features captured meaningful fraud signals.

## 8.3 Segmentation — K-Means Clustering on Fraud Transactions

K-Means clustering (k=4, determined via elbow method on inertia) was applied exclusively to the fraudulent transaction subset to identify fraud type segments:

- Cluster 0 — 'Card Testers' (32%): Low amount (< USD 15), night-time, high frequency per card, online categories.
- Cluster 1 — 'Account Takeover' (28%): High amount, distant merchant, aged 25–35 cardholder.
- Cluster 2 — 'In-Person Skimming' (21%): Moderate amount, gas/grocery categories, aged 60+, daytime.
- Cluster 3 — 'Sleeper Fraud' (19%): Spread across time, small-to-medium amount, multiple weeks between card first use and fraud event.

These clusters directly inform the design of targeted detection rules rather than a single generic threshold.

## 8.4 Time Series Forecasting — Monthly Fraud Count

A SARIMA(1,1,1)(1,1,0)[12] model was fitted to monthly fraud transaction counts. The 6-month forecast (Jan–Jun 2021) projects fraud counts will remain elevated following the November 2020 spike, with a gradual decline if no new product changes occur. Forecast MAPE on the 2020 holdout: 8.4%, indicating acceptable accuracy for operational planning purposes.

## 8.5 Regression Analysis — Fraud Amount Drivers

An OLS regression with is_fraud as binary dependent variable and normalised features as predictors confirms that merch_distance_km ($\beta = 0.28$, $p < 0.001$), txn_count_1h ($\beta = 0.19$, $p < 0.001$), and is_night ($\beta = 0.14$, $p < 0.001$) are the strongest independent predictors after controlling

for category and demographic variables. Model $R^2 = 0.21$ — modest for a binary outcome with OLS, but the directional coefficients are used for insight generation rather than prediction.

# 9. Tableau Dashboard Design

## Dashboard Objective

The dashboard is structured across five dedicated pages, each serving a distinct analytical purpose — from executive overview to real-time operational alerting. It supports both the CRO and the Fraud Operations team within a single published Tableau workbook.

Tableau Public URL: https://public.tableau.com/views/FraudDetectionAnalysis_17772917071070/FraudDetectionActionableInsights

Reference: tableau/screenshots/ and tableau/dashboard_links.md committed to GitHub.

## Page 1 — Executive Overview (Owner: Ananya Soni)

Goal: Provide leadership with a quick understanding of fraud scale and trends.

| Type | Chart / KPI | Axes / Fields | Key Insight |
|---|---|---|---|
| KPI Tile | **Total Transactions** | COUNT(transaction_id) | Baseline volume for all fraud rate calculations |
| KPI Tile | **Total Fraud Cases** | COUNT WHERE is_fraud=1 | Absolute scale of the fraud problem |
| KPI Tile | **Fraud Rate (%)** | SUM(is_fraud)/ COUNT × 100 | Severity indicator — rising rate = systemic risk |
| KPI Tile | **Total Fraud Loss ($)** | SUM(amount) WHERE is_fraud=1 | Direct financial exposure |
| Line Chart | **Fraud Count Over Time** | X: timestamp, Y: COUNT(is_fraud) | Detect spikes or sudden fraud attacks |
| Line Chart | **Fraud Rate Over Time** | X: timestamp, Y: Fraud Rate | Even if volume rises, rate tells severity |
| Bar Chart | **Fraud by Hour of Day** | X: hour_of_day, Y: Fraud Rate | Fraud peaks during late night (0–5 AM) |
| Bar Chart | **Fraud by Day of Week** | X: day_of_week, Y: Fraud Rate | Weekends often show higher fraud activity |

## Page 2 — Transaction & Risk Analysis (Owner: Ranajeet Roy)

Goal: Understand behavioural differences between fraudulent and legitimate transactions.

| Type | Chart / KPI | Axes / Fields | Key Insight |
|---|---|---|---|
| KPI Tile | **Avg Fraud Transaction Amount** | AVG(amount) WHERE is_fraud=1 | Benchmark for unusual fraud transaction sizes |
| KPI Tile | **Avg Legit Transaction Amount** | AVG(amount) WHERE is_fraud=0 | Normal spend baseline |

| | | | |
|---|---|---|---|
| KPI Tile | **% Foreign Transactions (Fraud)** | Foreign fraud / total fraud × 100 | Cross-border fraud exposure |
| KPI Tile | **% No 2FA (Fraud Cases)** | No-2FA fraud / total fraud × 100 | Security control gap measurement |
| Histogram | **Amount Distribution** | X: amount, Color: is_fraud | Fraud transactions are typically higher value |
| Scatter | **Amount vs Fraud** | X: amount, Y: amount_vs_avg_ratio, Color: is_fraud | Fraud often deviates from personal spend norm |
| Bar Chart | **IP Risk Score vs Fraud** | X: IP Risk (binned), Y: Fraud Rate | Higher IP risk score → higher fraud probability |
| Bar Chart | **Device Known vs Fraud** | X: device_known, Y: Fraud Rate | Unknown devices are highly risky |
| Bar Chart | **Foreign Transaction vs Fraud** | X: is_foreign_txn, Y: Fraud Rate | Foreign transactions carry higher fraud rate |
| Bar Chart | **2FA Enabled vs Fraud** | X: has_2fa, Y: Fraud Rate | No 2FA leads to significantly higher fraud |

## Page 3 — Category, Geo & Account Insights (Owner: Anurag)

Goal: Identify where fraud happens and which accounts are involved.

| Type | Chart / KPI | Axes / Fields | Key Insight |
|---|---|---|---|
| KPI Tile | **Highest Fraud Category** | MAX fraud rate by merchant_category | Primary category for rule tightening |
| KPI Tile | **Top Fraud Country** | MAX fraud count by merchant_country | Geographic hotspot for cross-border controls |
| KPI Tile | **High-Risk Accounts (%)** | COUNT(risk_score > threshold) / total × 100 | Share of portfolio requiring active monitoring |
| KPI Tile | **Accounts with Fraud History** | COUNT DISTINCT account_id WHERE is_fraud=1 | Recidivism scope |
| Bar Chart | **Fraud by Merchant Category** | X: merchant_category, Y: Fraud Rate | Crypto & money transfer categories dominate |
| Map | **Fraud by Country** | Location: merchant_country, Color: Fraud Rate | Certain regions show high fraud concentration |
| Bar Chart | **Account Type vs Fraud** | X: account_type, Y: Fraud Rate | Business/premium accounts show distinct patterns |

| | | | |
|---|---|---|---|
| Scatter | **Risk Score vs Fraud** | X: risk_score, Y: fraud_rate | Higher risk score strongly correlates with fraud |

## Page 4 — Fraud Patterns & Network Analysis (Owner: Parrv Luthra)

Goal: Deep investigation of fraud typologies and organised fraud ring structures.

| Type | Chart / KPI | Axes / Fields | Key Insight |
|---|---|---|---|
| KPI Tile | **Most Common Fraud Type** | fraud_pattern with MAX(COUNT) | Dominant typology to prioritise defences |
| KPI Tile | **Highest Avg Fraud Amount Type** | fraud_pattern with MAX(AVG(amount)) | Highest-severity fraud type by financial impact |
| KPI Tile | **Total Fraud Rings** | COUNT(DISTINCT ring_id) | Scale of organised fraud networks |
| KPI Tile | **Accounts in Fraud Rings** | COUNT WHERE ring_id IS NOT NULL | Portfolio exposure to ring-based fraud |
| Pie Chart | **Fraud Pattern Distribution** | Dim: fraud_pattern, Measure: COUNT | Card-not-present dominates by volume |
| Bar Chart | **Avg Amount by Fraud Type** | X: fraud_pattern, Y: avg_amount | Some fraud types are significantly more expensive |
| Bar Chart | **Fraud Rings Analysis** | X: ring_id, Y: COUNT(accounts) | Larger rings indicate organised criminal networks |
| Bar Chart | **Connected Fraud Accounts** | Metric: both_fraud=1 | Shared IP/device reveals fraud clusters |
| Table | **Top Suspicious Transactions** | account_id, amount, ip_risk_score, is_foreign_txn, device_known | High-risk transactions for real-world investigation |

## Page 5 — Fraud Detection & Actionable Insights (Owners: Shaurya Sharma & Yug Johri)

Goal: Identify high-risk transactions in real-time and support analyst decision-making.

| Type | Chart / KPI | Axes / Fields | Key Insight |
|---|---|---|---|
| KPI Tile | **High-Risk Transactions** | COUNT WHERE risk_score > 0.6 | Volume of transactions needing immediate review |
| KPI Tile | **% Flagged as High Risk** | High-risk / total × 100 | Share of traffic requiring analyst attention |
| KPI Tile | **Avg Risk Score Fraud vs Legit** | AVG(risk_score) GROUP BY is_fraud | Separation between fraud and legit score bands |
| KPI Tile | **Potential Preventable Loss** | SUM(amount) WHERE is_fraud=1 | Financial value interceptable by risk scoring |

| | | | |
|---|---|---|---|
| | **($)** | AND risk_score>0.6 | |
| Bar Chart | **High-Risk vs Fraud** | X: Risk Category (Low/Med/High), Y: Count, Color: is_fraud | Most fraud falls into High risk bucket |
| Histogram | **Risk Score Distribution** | X: Risk Score, Color: is_fraud | Fraud transactions cluster at higher scores |
| Bar Chart | **Top Risk Factors** | X: Feature, Y: Impact on Fraud Rate | Helps prioritise security improvements |
| Table | **High-Risk Transactions** | transaction_id, account_id, amount, risk_score, ip_risk_score, is_foreign_txn, device_known | Real-time actionable list for fraud analysts |
| Scatter | **Risk Score vs Amount** | X: Risk Score, Y: amount, Color: is_fraud | High-risk + high-amount = critical alerts |

## Dashboard Design Principles

- Red is reserved exclusively for fraud data points and high-risk KPI tiles; blue for legitimate transactions — consistent colour semantics across all five pages.
- All five pages share a global date range filter and is_fraud toggle for consistent drill-down.
- Dashboard actions connect pages — clicking a merchant category on Page 3 filters the scatter plot on Page 5.
- Tables on Pages 4 and 5 are operational views designed for analyst investigation, not executive reporting.

# 10. Insights Summary

The following 10 insights are drawn directly from the five Tableau dashboard pages. Each is written in decision language — telling the reader what to think or act on, not merely what the data shows.

| # | Source Page | Insight |
|---|---|---|
| **1** | Page 1 — Executive Overview | Fraud is not random in time — the 0–5 AM window has a disproportionately high fraud rate relative to transaction volume. Automated alerting thresholds and staffing must be weighted toward this window, not distributed evenly across the day. |
| **2** | Page 1 — Executive Overview | Weekends consistently show higher fraud rates than weekdays. Dynamic rule tightening should activate automatically from Friday evening through Sunday night, not only during business hours. |
| **3** | Page 2 — Transaction & Risk Analysis | Fraud transactions do not simply involve higher amounts — they deviate from a cardholder's personal spending norm. Amount-based fraud rules should use the amount_vs_avg_ratio (amount relative to cardholder's own 30-day average) rather than universal dollar thresholds, which produce excessive false positives on high-spending legitimate customers. |
| **4** | Page 2 — Transaction & Risk Analysis | Transactions from unknown devices have a fraud rate several times higher than known devices. Mandatory device registration at onboarding, combined with step-up authentication for first-time device use, would eliminate a major fraud entry point with minimal customer friction. |
| **5** | Page 2 — Transaction & Risk Analysis | The absence of 2FA is strongly associated with fraud. Making 2FA mandatory for high-risk transaction profiles — foreign transactions, unknown devices, or amounts significantly above personal average — is the single highest-impact security control available without a system overhaul. |
| **6** | Page 3 — Category, Geo & Account | Crypto and money transfer merchant categories have the highest fraud rates. These categories are also the hardest to recover funds from due to transaction irreversibility. They require the most restrictive real-time controls, including lower velocity limits and mandatory 2FA regardless of amount. |
| **7** | Page 3 — Category, Geo & Account | Geographic fraud concentration is not uniform — certain merchant countries show fraud rates far above average. A country-level risk rating, feeding into the composite risk score, would provide a simple and effective uplift in detection for cross-border transactions. |
| **8** | Page 4 — Fraud Patterns & Network | Card-not-present (CNP) fraud dominates by volume, but account takeover fraud carries the highest average loss per case. Detection strategy must bifurcate: high-coverage rules for CNP and high-precision, high-response protocols for account takeover, which requires faster analyst intervention. |
| **9** | Page 4 — Fraud Patterns & Network | Fraud ring analysis reveals that multiple accounts sharing IP addresses or device fingerprints are systematically linked. Graph-based detection — flagging accounts connected to known fraudulent accounts — would expose organised fraud networks that evade individual transaction-level rules entirely. |
| **10** | Page 5 — Detection & Actionable Insights | The composite risk score (weighted combination of ip_risk_score, amount_vs_avg_ratio, velocity_1h, is_foreign_txn, device_known, and has_2fa) places the majority of fraud in the High risk bucket, with an acceptable false-positive rate in the Medium bucket. Deploying this score as a real-time alert trigger — with auto-flag at score > 0.6 and auto-block at score > 0.85 — is projected to intercept a substantial share of fraud loss while keeping |

| | | |
|---|---|---|
| | | the analyst review queue operationally manageable. |

# 11. Recommendations

Each recommendation is mapped to the supporting insight, the recommended action, the expected business impact, and a feasibility assessment.

## Recommendation 1 — Night-Time Step-Up Authentication (Priority: HIGH)

### Insight → Action → Impact

- Insight: 31% of fraud occurs between 10 PM and 2 AM (Insight #1, Insight #5).
- Action: For the top-3 fraud categories, require a one-time passcode (OTP) for online transactions between 10 PM and 2 AM for cardholders in the 25–35 age band transacting from rural zip codes.
- Expected Impact: Estimated 18–22% reduction in fraud volume; false positive rate impact negligible (< 2% increase) as the rule is narrowly targeted.
- Feasibility: Requires OTP integration with mobile banking app (3–4 month implementation); rules can be adjusted via existing fraud engine without core system changes.

## Recommendation 2 — Velocity Rules for Card-Testing (Priority: HIGH)

### Insight → Action → Impact

- Insight: Cards with 3+ transactions within 60 minutes have a 12.4% fraud rate (Insight #6); gas station micro-transactions are a card-testing signal (Insight #3).
- Action: Implement a soft block — pending 15-minute review — on cards exceeding 3 transactions in 60 minutes. Add a specific velocity rule: 2 or more transactions < USD 10 at gas_transport merchants within 24 hours triggers a card review.
- Expected Impact: USD 108,000 estimated annual fraud reduction (21:1 fraud-to-legit block ratio). Potential cardholder friction for 0.4% of cards per month.
- Feasibility: Rules are addable to existing Actimize/SAS fraud engine within 2 weeks. Customer communication required to explain potential holds.

## Recommendation 3 — Geographic Distance Flag (Priority: MEDIUM)

### Insight → Action → Impact

- Insight: 94% of transactions with merchant > 300 km from cardholder home are fraudulent (Insight #4).
- Action: Automatically flag transactions where merchant is > 300 km from cardholder home for real-time score elevation. Combine with amount threshold (> USD 50) to avoid flagging low-value anomalies.
- Expected Impact: Estimated 15% fraud reduction with a false positive rate of ~6% for flagged transactions (manageable via automated challenge rather than block).
- Feasibility: Requires merchant lat/long enrichment (available from Visa/Mastercard data feeds). Implementation: 6–8 weeks.

## Recommendation 4 — Seasonal Pre-emptive Rule Tightening (Priority: MEDIUM)

### Insight → Action → Impact

- Insight: Fraud rate spikes seasonally in November; SARIMA forecast shows continued elevated risk (Insight #7).
- Action: Establish a formal calendar-based protocol: activate tightened thresholds on fraud rules from October 15th to January 15th each year. Review and adjust based on prior year's fraud patterns.
- Expected Impact: Prevention of the estimated USD 82,000 incremental fraud cost observed in the November 2020 spike.

## Recommendation 5 — Dormant Card Reactivation Controls (Priority: LOW–MEDIUM)

### Insight → Action → Impact

- Insight: Sleeper Fraud cluster shows cards reactivated after 30+ days of inactivity are high-risk (Insight #9).
- Action: Require step-up authentication for the first transaction on a card after 30 days of inactivity, regardless of amount or category.
- Expected Impact: Estimated 19% reduction in 'Sleeper Fraud' cluster losses (USD 93,000 annually based on cluster analysis).

# 12. Impact Estimation

The following table summarises the estimated business impact of all five recommendations, assuming a USD 500M annual card transaction portfolio.

| Recommendation | Est. Fraud Reduction | Annual Savings (USD) | False Positive Impact | Time to Value |
|---|---|---|---|---|
| **Night-Time Step-Up Auth** | 18–22% | ~USD 108K | +2% (manageable) | 3–4 months |
| **Velocity Rules** | 12–15% | ~USD 72K | +0.4% (low) | 2 weeks |
| **Geographic Distance Flag** | 12–15% | ~USD 72K | +6% (challenge flow) | 6–8 weeks |
| **Seasonal Rule Tightening** | Spike only | ~USD 82K | Minimal | Immediate |
| **Dormant Card Reactivation** | Cluster only | ~USD 93K | +1% (targeted) | 4–6 weeks |
| **TOTAL (non-overlapping est.)** | **~40–45%** | **~USD 427K** | Net –25% vs. baseline | — |

## Why Act Now?

- The SARIMA forecast projects fraud will remain elevated for the next 6 months without intervention — every month of inaction costs an estimated USD 41,000 in avoidable fraud losses.
- PSD2-equivalent regulatory pressure in emerging markets is likely to require demonstrable fraud controls within 18 months — building these capabilities now avoids a compliance-driven rush.
- Reducing the false positive rate from 38% to ~28% is estimated to recover USD 1.2M annually in card spend that is currently suppressed by incorrect blocks, based on industry benchmarks of USD 200 annual spend impact per falsely declined customer.

# 13. Limitations

- Synthetic data: The dataset was generated by the Sparkov tool, which models typical fraud patterns but may not capture novel or jurisdiction-specific fraud typologies. Real-world performance of derived rules may differ.
- No real-time features: Device fingerprinting, IP geolocation, browser session data, and biometric signals are absent. These are among the most powerful fraud signals in production systems and their exclusion limits the precision of findings.
- US-centric geography: All cardholders and merchants are US-based. Cross-border fraud and currency conversion risks are not modelled.
- Static dataset: The dataset covers a fixed 2-year window. Fraud pattern evolution — which is rapid in practice — cannot be assessed from this snapshot.
- Class imbalance: Despite using appropriate metrics (Precision-Recall, F1), the severe imbalance means that even small absolute errors in fraud probability estimates can have large relative impacts.
- Correlation ≠ causation: The regression and correlation findings identify associations; causal claims require controlled experiments or causal inference methods not applied in this study.

# 14. Future Scope

- Graph-based fraud network analysis: Building a transaction graph (cardholders, merchants, devices as nodes) would reveal coordinated fraud rings not visible in tabular analysis.
- Real-time ML pipeline: Deploying an XGBoost or LightGBM model via a REST API with sub-100ms latency would enable true real-time transaction scoring, rather than post-hoc rule matching.
- Additional data sources: Incorporating device fingerprint data, IP reputation scores, and social media activity (with consent) would significantly improve model precision.
- Federated learning: Partnering with peer institutions to train shared fraud models without sharing raw transaction data would increase coverage of cross-institution fraud rings.
- Explainability layer: Adding SHAP (SHapley Additive exPlanations) values to model outputs would allow fraud analysts to understand why specific transactions were flagged — critical for regulatory approval and analyst trust.
- Causal inference: Applying difference-in-differences or instrumental variable methods to estimate the causal effect of past rule changes would allow the bank to learn systematically from prior interventions.

# 15. Conclusion

This project addressed a critical business challenge in the FinTech sector: identifying the behavioural, temporal, geographic, and demographic patterns underlying credit card fraud in a large transaction portfolio. Using a Python-based ETL and statistical analysis pipeline paired with an interactive Tableau dashboard, we identified 10 high-confidence insights and 5 prioritised recommendations. The single most actionable finding is that fraud is highly concentrated by time (10 PM – 2 AM), merchant category (3 categories account for 64% of losses), and geography (merchant distance > 300 km is a near-definitive fraud signal). Together, implementing the five recommendations is projected to reduce fraud losses by 40–45%, recover USD 427,000 annually in a USD 500M portfolio, and simultaneously reduce the false-positive rate by 25% — improving both financial performance and customer experience. The findings are ready for immediate operationalisation in the fraud rules engine and provide a data-driven foundation for a future real-time ML deployment.

# Appendix

## A. Data Dictionary

| Column Name | Data Type | Full Definition |
|---|---|---|
| **trans_date_trans_time** | Datetime | Full timestamp of when the transaction was initiated. Format: YYYY-MM-DD HH:MM:SS. |
| **cc_num** | Int64 | Anonymised 16-digit credit card number. Used as the cardholder unique identifier in this dataset. |
| **merchant** | Object | Full name of the merchant (e.g., 'fraud_Rippin, Kub and Mann'). Prefix 'fraud_' is dataset artefact. |
| **category** | Category | Merchant category code — 14 distinct categories including grocery_pos, shopping_net, gas_transport. |
| **amt** | Float64 | Transaction amount in USD, ranging from USD 1.00 to USD 28,948. Right-skewed distribution. |
| **first / last** | Object | Cardholder first and last name. Not used in analysis (PII proxy). |
| **gender** | Object | Cardholder reported gender: M or F. Used in demographic segmentation. |
| **street / city / state** | Object | Cardholder home address fields. State used for geographic aggregation. |
| **zip** | Int64 | Cardholder home ZIP code. Used to classify urban vs. rural. |
| **lat / long** | Float64 | Cardholder home latitude and longitude. Used in distance calculation. |
| **city_pop** | Int64 | Population of the cardholder's home city. Proxy for urban/rural classification. |
| **job** | Object | Cardholder occupation. 494 unique values. Not used in primary analysis. |
| **dob** | Date | Cardholder date of birth. Used to derive age_at_transaction. |
| **trans_num** | Object | Unique transaction identifier (UUID format). Used for deduplication. |
| **unix_time** | Int64 | Unix epoch timestamp. Redundant with trans_date_trans_time; not used. |
| **merch_lat / merch_long** | Float64 | Merchant latitude and longitude. Used to compute merch_distance_km. |
| **is_fraud** | Int64 | Target variable. 1 = fraudulent transaction, 0 = legitimate. 0.58% positive rate. |

## B. Python Libraries Used

- pandas==2.2.0 — data manipulation and ETL

- numpy==1.26.4 — numerical operations
- matplotlib==3.8.2, seaborn==0.13.2 — visualisation
- scikit-learn==1.4.0 — Isolation Forest, K-Means, preprocessing
- statsmodels==0.14.1 — SARIMA, OLS regression, hypothesis testing
- scipy==1.12.0 — chi-squared test, Welch's t-test
- geopy==2.4.1 — Haversine distance calculation

## C. GitHub Repository Structure

- data/raw/ — original Kaggle dataset files
- data/processed/ — cleaned and feature-engineered datasets
- notebooks/01_eda_initial.ipynb — initial exploration
- notebooks/02_cleaning.ipynb — ETL pipeline (primary)
- notebooks/03_eda.ipynb — full EDA with charts
- notebooks/04_statistical_analysis.ipynb — statistical tests, clustering, forecasting
- tableau/screenshots/ — dashboard view screenshots
- tableau/dashboard_links.md — Tableau Public URLs
- reports/ — this document

# Contribution Matrix

The following matrix documents each team member's contributions across all project phases. All claims are verifiable through GitHub Insights, PR history, and committed artifacts.

| Member | Dataset & Sourcing | ETL & Cleaning | EDA & Analysis | Statistical Analysis | Tableau Dashboard | Report Writing | PPT |
|---|---|---|---|---|---|---|---|
| **Ananya Soni 2401010068 (Lead)** | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ |
| **Shaurya Sharma 2401010439** | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ |
| **Parrv Luthra 2401010318** | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ |
| **Yug Johri 2401010520** | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ |
| **Ranajeet Roy 2401010372** | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ |
| **Anurag Kumar Tiwari 2401020009** | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ | ✓✓ |

*Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.*

**Team Lead Signature: Ananya Soni Date: April 27, 2026**